

Text Analytics: Topic Modeling, Vocabulary Structure, and Document Embeddings

This section examines the latent semantic structure of the Active Inference corpus through complementary text-analytic methods: non-negative matrix factorization for topic discovery, TF-IDF vocabulary analysis, document embedding projections, and term co-occurrence patterns. Together, these analyses reveal thematic structure that cuts across the keyword-based domain taxonomy presented in the field overview.

Topic Modeling: Latent Structure

Non-negative matrix factorization (NMF) applied to the TF-IDF matrix identifies five latent topics:

Table 1: Non-negative matrix factorization (NMF) topic decomposition of the corpus TF-IDF matrix ($k = 5$ topics). Top terms are ranked by NMF component weight; interpretations reflect dominant thematic content.

Topic	Top Terms
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0	learning, agent, model, agents, active, environments, aif, inference
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